

Fm 16 Leon McQueen Vectors 2

$$1 \quad OA = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

$$OB = \begin{pmatrix} 8 \\ -3 \\ 2 \end{pmatrix}$$

$$|OA| = \sqrt{2^2 + 3^2 + 5^2}$$

$$= \sqrt{4 + 9 + 25}$$

$$= \sqrt{38}$$

$$\therefore \text{unit vector} = \frac{1}{\sqrt{38}} OA$$

$$= \begin{pmatrix} \frac{2}{\sqrt{38}} \\ \frac{3}{\sqrt{38}} \\ \frac{5}{\sqrt{38}} \end{pmatrix}$$

$$2 \quad AO = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$

$$OB = \begin{pmatrix} -1 \\ 4 \\ 8 \end{pmatrix}$$

$$AB = AO + OB$$

$$= \begin{pmatrix} -4 \\ 2 \\ 9 \end{pmatrix}$$

$$|AB| = \sqrt{(-4)^2 + 2^2 + 9^2}$$

$$= \sqrt{101}$$

$$3 \quad A = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} \quad C = \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}$$



$$BC = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$AD = BC$$

$$\text{Let } D = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$AD = \begin{pmatrix} x+1 \\ y-3 \\ z-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$$

$$x+1=4$$

$$x=3$$

$$y-3=1$$

$$y=4$$

$$z-2=0$$

$$z=2$$

$$\therefore D = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$$

$$4) BC=AD$$

$$B C = \begin{pmatrix} -4 \\ 3 \\ -3 \end{pmatrix}$$

$$AD = D - \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$= D + \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

$$D + \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ -3 \end{pmatrix}$$

$$D = \begin{pmatrix} -1 \\ 4 \\ -5 \end{pmatrix}$$

$$5) |AC| = |C - A|$$

$$= \left| \begin{pmatrix} 9 \\ 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \right|$$

$$= \left| \begin{pmatrix} 6 \\ 2 \\ 3 \end{pmatrix} \right|$$

$$= \sqrt{36 + 4 + 9}$$

$$= \sqrt{49}$$

$$6a) A = \begin{pmatrix} 8 \\ 6 \\ 0 \end{pmatrix} \quad C = \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$$

$$AC = \begin{pmatrix} 2-8 \\ 4-6 \end{pmatrix}$$

$$AB = \begin{pmatrix} 4-8 \\ 2-6 \\ 0 \end{pmatrix}$$

$$|AC| = |AB|$$

$$\therefore (2-a)^2 + (4-b)^2 + 2^2 = (a-8)^2 + (2-b)^2$$

$$a^2 - 4a + 4 + b^2 - 8b + 16 + 4 = a^2 - 16a + 64 + b^2 - 4b + 4$$

$$a^2 - 4a + b^2 - 8b + 24 = a^2 - 16a + b^2 - 4b + 20$$

$$4a - 4b + 4 = 0$$

$$a - b + 1 = 0$$

$$b) a - b + 1 = 0$$

$$a = b - 1$$

$$\therefore A = \begin{pmatrix} 8 \\ 6 \\ 0 \end{pmatrix}$$

$$\text{Min area} = \text{minimize } |AB| = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 8 \\ 6 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ -4 \\ 0 \end{pmatrix}$$

$$|AB|^2 = (5b)^2 + (2-b)^2$$

$$= 26^2 - 14b + 2b^2$$

Min point: $46 - 14 = 0$
 $b = \frac{7}{2}$

$\therefore$

$$A = \frac{1}{2} \begin{pmatrix} \frac{5}{2} \\ \frac{7}{2} \\ 0 \end{pmatrix} \quad \begin{pmatrix} \frac{2}{2} \\ \frac{5}{2} \\ 0 \end{pmatrix}$$

$$7 \quad A = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$$

$$C = \begin{pmatrix} 4 \\ 5 \\ -2 \end{pmatrix}$$

$$a) \quad AB = \begin{pmatrix} 2 \\ -7 \\ 7 \end{pmatrix}$$

$$BD = AB$$

$$D - B = \begin{pmatrix} 2 \\ 7 \\ 7 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 \\ 7 \\ 7 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 6 \\ 10 \\ 10 \end{pmatrix}$$

$$b) \quad AC = \begin{pmatrix} a-2 \\ 2 \\ 2 \end{pmatrix}$$

$$|AC| = \sqrt{(a-2)^2 + 2^2 + 2^2}$$

$$= a^2 - 4a + 4 + 4 + 4$$

$$= \sqrt{a^2 - 4a + 12} = 4$$

$$a^2 - 4a + 12 = 16$$

$$a^2 - 4a - 4 = 0$$

$$a = \frac{4 \pm \sqrt{16 + 16}}{2}$$

$$= \frac{4 \pm 4\sqrt{2}}{2}$$

$$a < 0 \therefore a = \frac{4 - 4\sqrt{2}}{2}$$

$$= 2 - 2\sqrt{2}$$

$$8 \ A = \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix}$$

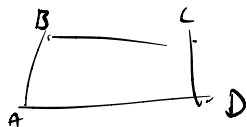
$$B = \begin{pmatrix} 3 \\ -3 \\ -4 \end{pmatrix}$$

$$C = \begin{pmatrix} 2 \\ -6 \\ 4 \end{pmatrix}$$

$$a) \begin{pmatrix} 3 \\ -3 \\ -4 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix} = \begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$$

$$b) \ AB = \begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$$

$$CD = \begin{pmatrix} 2 \\ -6 \\ 4 \end{pmatrix} = 2AB$$



$\therefore$ same gradient

$\therefore$ parallel

so trapezium! two parallel sides.

$$9) \ D = \begin{pmatrix} -1 \\ 7 \\ 3 \end{pmatrix}$$

b) AB & DC must be parallel

$$AB = \begin{pmatrix} 6 \\ -3 \\ p-7 \end{pmatrix}$$

$$DC = CB + BO + OD$$

$$= \begin{pmatrix} 9 \\ -9 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ 1 \\ 7 \end{pmatrix} + \begin{pmatrix} -7 \\ 3 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ 2 \\ 10-p \end{pmatrix}$$

$$\therefore AB = k DC$$

$$\begin{pmatrix} 6 \\ -3 \\ p-7 \end{pmatrix} = k \begin{pmatrix} -4 \\ 2 \\ 10-p \end{pmatrix}$$

$$\text{By hypothesis, } k = \frac{3}{2}$$

$$\frac{3}{2}(10-p) = p-7$$

$$3(10-p) = 2p-14$$

$$3p-30 = 2p-14$$

$$p = 16$$

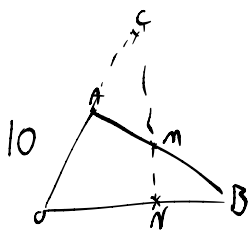
$$9) \ AC = \begin{pmatrix} 3 \\ -6 \\ 4 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 \\ -8 \\ 2 \end{pmatrix}$$

$$\text{Let } \theta = \angle BAC$$

$$\cos \theta = \frac{3 \times 1 + 3 \times -6 + 4 \times 2}{\sqrt{(3^2 + 3^2 + 4^2)} \sqrt{(1^2 + 8^2 + 2^2)}} \\ = -0.27375$$

$$\arccos(-0.27375) = 105.88^\circ \\ = 105.9^\circ$$



$$AB = b - a$$

$$OA = a$$

$$\therefore OM = \frac{1}{2}(b-a) + a$$

$$= \frac{1}{2}(a+b)$$

$$OC = 2a$$

$$CO = -2a$$

$$\therefore CM = \frac{1}{2}a - 2a + \frac{1}{2}b$$

$$a) = -\frac{3}{2}a + \frac{1}{2}b$$

$$b) CN = \lambda CM$$

$$\therefore ON = 2a + \lambda CM$$

$$= 2a + \frac{3}{2}\lambda a + \frac{1}{2}\lambda b$$

$$= (2 + \frac{3}{2}\lambda)a + \frac{1}{2}\lambda b$$

$$ON = \phi OB = \phi b$$

$$\therefore 2 + \frac{3}{2}\lambda = 0$$

$$2 = -\frac{3}{2}\lambda$$

$$4 = -3\lambda$$

$$\lambda = -\frac{4}{3}$$

$$\therefore \frac{1}{2}\lambda b = -\frac{2}{3}b$$

$$= -\frac{2}{3}b$$

$$\therefore \phi = -\frac{2}{3}$$

$$\therefore ON : NB = \frac{2}{3} : \frac{1}{3} = 2 : 1$$